

Scattering

Applied aspects-Glare effect-light reduction effect

Photo electric effect, Raman Effect and LASER

Glare effect - Introduction

- Light is crucial for vision.
- It bounces off objects and enters your eyes, which allows you to see.
- But sometimes, it's the source of vision problems, like glare.
- The issue of glare is something most people deal with on a daily basis
- The effects of glare can become an unpleasant

Glare – light reduction effect

- Glare is the **loss of visual performance** or **discomfort** produced by an **intensity of light** in the visual field **greater than the intensity of light to which the eyes are adapted**.
- In simple, glare occurs when too much light enters the eye and interferes with the eye's ability to manage it.
- Glare can be described as “extreme brightness” from the presence of excessive visible light, and can cause discomfort and the vision to be compromised
- Glare can be distracting and even dangerous and can occur day or night in several ways.
- Glare may come directly from a light source or be reflected.

Types of glare

- There are four types of glare:
 - Distracting glare,
 - discomforting glare,
 - disabling glare, and
 - blinding glare

- **Distracting glare** – Distracting glare results from light being **reflected** when it moves from **one optical medium to another** (e.g. from air to glass).
- This type of glare can also be experienced at night, forming “halos”(Seeing bright circles or rings around a light source, around headlights or streetlights, are known as **halos**).
- Distracting glare can represent an annoyance or distraction to the viewer and lead to **eye fatigue**.



- **Discomforting glare** – Discomforting glare may result from **direct or reflected** glare and can be caused by everyday **bright sunlight** conditions.
- The unprotected eye will respond to discomforting glare by **squinting and constriction** of the pupil.
- It is common for the affected individual to try to avoid the glare by **shielding the eyes or turning another direction**.

- **Disabling glare** – Disabling glare, also known as veiling glare, is **more intense than discomforting glare** and the **high level of light** produces a glare that can actually **interfere or block vision**.
- This type of glare comes from **excessive, intense light** that can occur when you face **directly into the sun**
- Disabling glare tends to become more **problematic in the elderly**, as the decreasing transparency of the crystalline lens that comes with age leads to developing cataract formation.



- **Blinding glare** – Blinding glare results from **light reflecting off of smooth, shiny surfaces such as water, sand or snow.**
- It can be **strong enough to block vision.** When the light reflects off the surfaces, it becomes polarized and produces blinding glare.
- Glare tends to be worse with old age, light colored eyes, prior eye surgery, and **certain eye conditions**, including **cataracts, dry eye** and **macular degeneration.**
- Macular degeneration is **deterioration** of the macula, which is the small central area of the retina of the eye that is responsible for sharp, central vision

Glare and Halos

- **Glare** is light that enters the eye and hinders the vision, like when a camera flash goes off.
- Seeing bright circles or rings around a light source, like headlights, are known as **halos**.
- Halos around lights are most noticeable at night or when the person is in dim or dark areas.

What Causes Them?

- **Cataracts**
- Normally, the lens at the front of the eye is clear.
- Light can pass through easily.
- A **cataract scatters rather than focuses light.**
- This blurs the vision.
- Halos are a common symptom. Glare might make you think that lights are too bright.
- **Common eye problems** . retina is the thin lining in the back of the eye.
- It plays a crucial role in vision. If light can't focus on it, we may start to see halos or glare.

- **Conditions that can cause this include:**
 - **Nearsightedness** (hard to see things that are far away, often worse at night)
 - **Farsightedness** (hard to see things nearby due to the natural shape of your eyeball)
 - **Astigmatism** (blurred vision due to irregular shape of the cornea, the front surface of the eye)

What optical solutions exist to reduce glare?

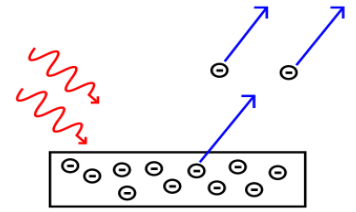
Some simple steps to reduce glare:

- **Sunglasses** : They can help during the day. Polarized shades can keep the eyes safe from certain types of glare, like reflections from water.
- **Vehicle visor.** Use this to keep direct sunlight out of your eyes.
- **Glasses or contact lenses** : special types of lenses that can help reduce glare and correct eye problems.

- **Polarized Lenses**

- These reduce glare at the beach, in the snow, or out on the water. But they don't take the place of UV protection.
- You might see better through them when there's tons of light around.
- But they can make it harder to see things like computer screens, smartphones, or dashboards.

Photoelectric effect



- The **photoelectric effect** is the emission of electrons or other free carriers when light hits a material.
- Electrons emitted in this manner can be called *photoelectrons*.
- Applications of the photoelectric effect brought us "electric eye" door openers, light meters used in photography, solar panels and photostatic copying.

Applications of photoelectric effect

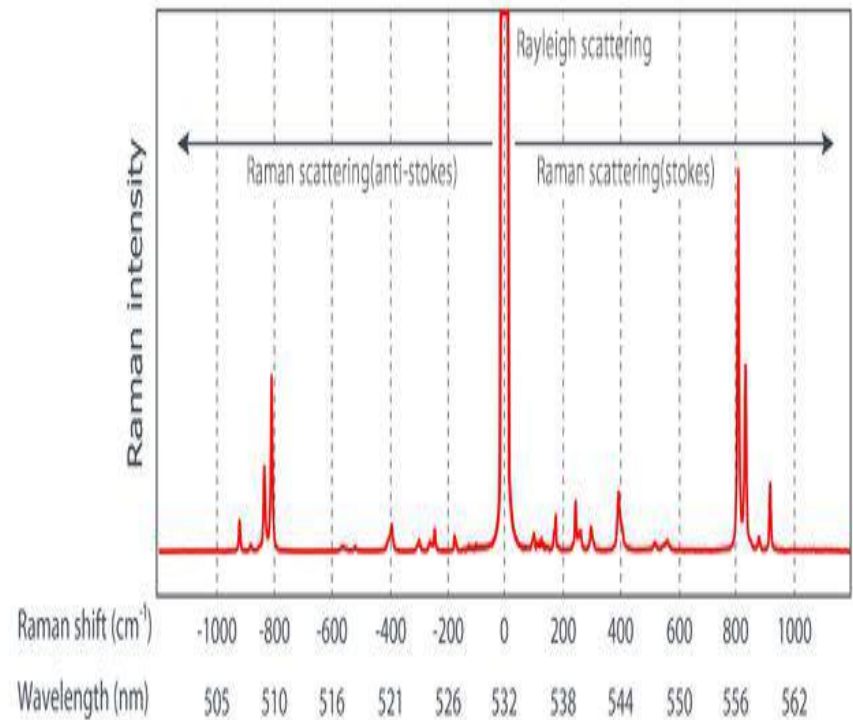
- 1. **Photo - cell** is the most important application . It is most commonly found in solar panels. It works on the basic principle of the light striking the cathode which causes the emission of electrons, which in turn produces current.
- An electric eye is a photodetector used for detecting obstruction of a light beam.
- 2. **Photomultiplier tube** makes use of photo electric effect to convert light intensity into electrical currents
- 3. Photoelectric effect also finds application in **photocopies, light meter, photodiodes, phototransistors.**
- 4. **Scintillators** : A scintillator is a device that emit light when it attracts radiation from either source in the lab or a cosmic sources

Raman scattering

- When light is irradiated on molecules, the light is scattered by molecules.
- Most of scattered light have the same frequency with incident light but some fraction of light have **different frequency** due to interaction between oscillation of light and molecular vibration.
- **The phenomenon that the light is scattered with frequency change is called Raman scattering.**
- Because this frequency modulation is specific to molecular vibration, it is possible to analyze composition of material or crystal lattice information by analyzing spectrum of Raman scattered light.
- This analysis method is Raman spectroscopy.

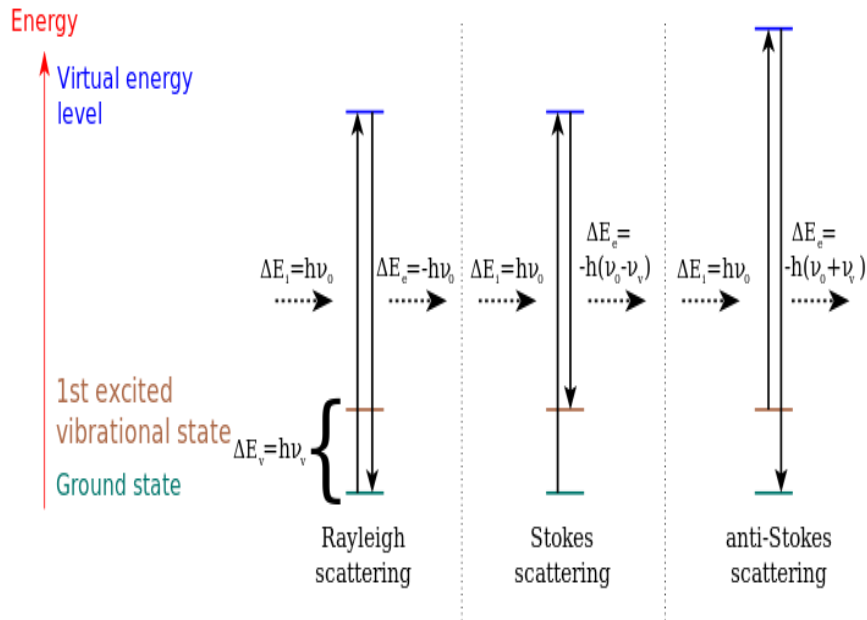
Raman spectrum

- Raman scattered light contains various information on molecules in a substance.
- If you separate the scattered light by wavelength in order to interpret them, you can see that Rayleigh scattered light equal to the wavelength of the incident light is strongly detected and Raman scattered light is detected on both sides as shown in the above figure.
- Raman scattered light detected on the **shorter wavelength side** then Rayleigh scattered light is called **anti-Stokes line**, and that detected on the **longer wavelength side** is called **Stokes line**. In general, Stokes lines with higher intensity are used for analysis.



Stokes and anti-Stokes

- The Raman interaction leads to two possible outcomes:
- the material absorbs energy and the **emitted photon** has a **lower energy than the incident photon**.
- This outcome is labeled **Stokes Raman** scattering in honor of George Stokes who showed in 1852 that fluorescence is due to light emission at longer wavelength (now known to correspond to lower energy) than the absorbed incident light.
- the material loses energy and the **emitted photon** has a **higher energy than the absorbed photon**.
- This outcome is labeled **anti-Stokes** Raman scattering.
- The energy difference between the **absorbed and emitted photon** corresponds to the **energy difference between two resonant states** of the material and is independent of the absolute energy of the photon.
- It is used with **microscopes**, typically in combination with near-infrared (NIR) laser excitation



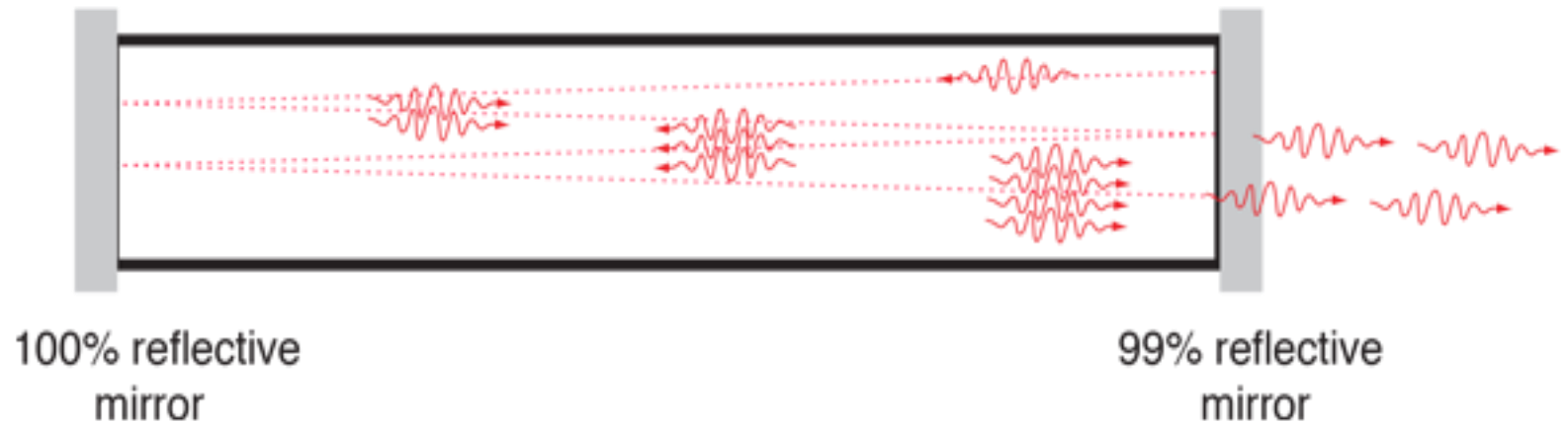
- The different possibilities of light scattering:
- Rayleigh scattering (no exchange of energy: incident and scattered photons have the same energy),
- Stokes Raman scattering (atom or molecule absorbs energy: scattered photon has less energy than the incident photon) and
- anti-Stokes Raman scattering (atom or molecule loses energy: scattered photon has more energy than the incident photon)

Raman Scattering Applications

- **Polymorphs:** Polymorphism occurs when a molecule is able to exist in **more than one crystalline form** . Raman spectroscopy is ideal for the **recording the differences in the two forms** and in **measuring the forms** while optimizing and during the crystallization process.
- **Polymerization:** Raman spectroscopy tends to provide a stronger signal (than IR) from the molecular backbone, especially double and triple carbon bonds. For this reason Raman can be a better choice for **identifying polymers and in monitoring polymerization reactions**
- **Crystallization:** Crystalline materials can be **characterized and distinguished according to the structure of their unit cell**. Many crystalline materials can **form different polymorphs** in order to minimize their **crystal lattice energy** under specific thermodynamic conditions. Molecules exhibit "fingerprint" vibrational patterns and Raman is able to **identify different polymorphs by their characteristic fingerprints**. These differences may be simply the distance or angle between bonds, nevertheless producing a different Raman fingerprint

LASER

Light **A**mplification by **S**timulated **E**mission of **R**adiation



LASER

- A laser is a device which produces highly directional light.
- Laser is an acronym for **Light Amplification by Stimulated Emission of Radiation**.
- From the term, it appears that it is a device which amplifies light radiation which is obtained by stimulated emission process
- The mechanism by which this is accomplished, stimulated emission, was first postulated by Albert Einstein in 1917
- The first laser was built in 1960 by Theodore H. Maiman at Hughes Research Laboratories, based on theoretical work by Charles Hard Townes and Arthur Leonard Schawlow

Characteristics

- Monochromaticity
- High Intensity
- Coherence
- Directionality



Properties of LASER

- Laser light is **monochromatic, bright, unidirectional and coherent**
- **Monochromaticity** means luminous waves emitted come out with **same wavelength and energy**.
- **Brilliancy** (high intensity) means light beam emitted is **extremely intense** and angularly well centered.
- **Coherency** means all the photons emitted vibrate **in phase** agreement both in space and time.
- **Unidirectional** means all the photons travel in a **single direction**.

....Properties of LASER

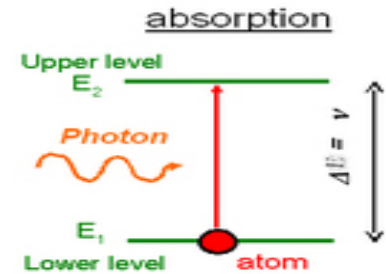
- Light from laser has very low divergence.
- The laser beam is very narrow and can be concentrated on a very small area. This makes laser light highly directional.
- Hence it can travel over great distances or can be focused to a very small spot with brightness greater than the sun
- The light waves of ordinary light sources have many wavelengths.
- Hence, the photons emitted by ordinary light sources are out of phase.
- Thus, ordinary light is incoherent.

Laser principle

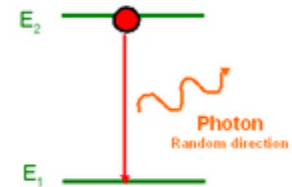
- Laser principle can be understood from Bohr's model.
- Light is made up of particles called "photons". Each photon has energy (E) expressed as follows.
 - ➡ $E = h\gamma$, where h is planck's constant and γ is frequency of light.
 - ➡ $\lambda \gamma = c$
 - ➡ $E = hc/\lambda$

Principles of working of a laser

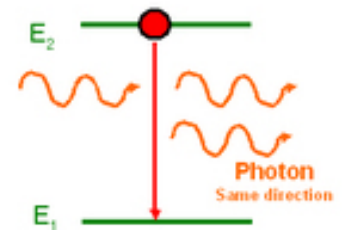
- In lasers, photons are interacted in three ways with the atoms:
 - Absorption of radiation
 - Spontaneous emission
 - Stimulated emission
- **Absorption** of radiation is the process by which electrons in the ground state absorb energy from photons to jump into the higher energy level.
- **Spontaneous emission** is the process by which electrons in the excited state return to the ground state by emitting photons
- **Stimulated emission** is the process by which incident photon interacts with the excited electron and **forces** it to return to the ground state



Spontaneous emission



Stimulated emission

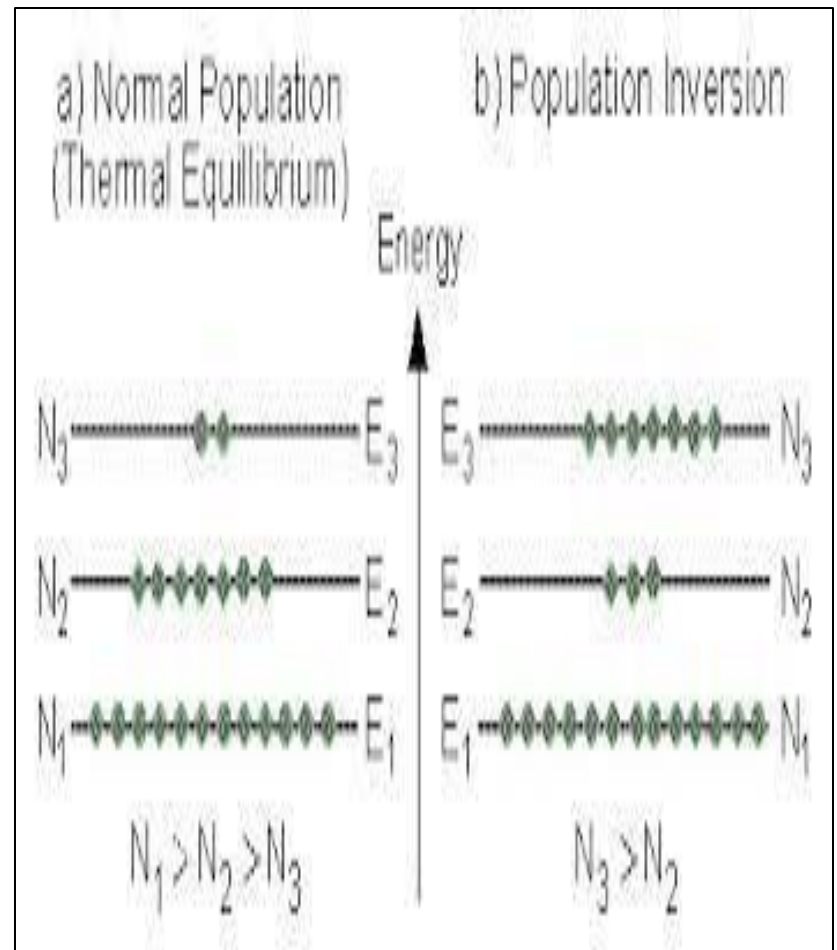


Definition - Population Inversion

- Population inversion is the process of achieving greater population of higher energy state as compared to the lower energy state. This condition is called population inversion
- Population inversion technique is mainly used for light amplification by stimulated emission of radiation
- The population inversion is required for laser operation.

Population inversion and pumping

- If there are more atoms in the upper level (N_3) than in the lower level (N_2), the system is not at equilibrium.
- In fact, at thermodynamic equilibrium, the distribution of the atoms between the levels is given by Boltzmann's Law:
- A situation not at equilibrium must be created by adding energy via a process known as “pumping” in order to raise enough atoms to the upper level.



- This is known as population inversion and is given by $\Delta = N_3 - N_2$
- Light is amplified when the population inversion is positive.
- Pumping may be electrical, optical or chemical.

Pumping mechanisms

- The process of sending atoms from lower energy state to higher energy state is called Pumping.
- Optical pumping
- Electric discharge
- Chemical reaction
- Injection current through p-n junction

Optical Feed back

- To direct the amplified light to travel back and forth through the active medium many times two end mirrors are kept at both the ends of the laser. These mirrors provide necessary optical feed back.

Types of lasers

- There are many types of lasers available for research, medical, industrial, and commercial uses. Lasers are often described by the kind of lasing medium they use –
- solid state,
- gas,
- excimer,
- dye, or liquid
- semiconductor.

- **Solid State lasers** : **Solid-state lasers** have lasing material distributed in a solid matrix (such as the ruby or neodymium:yttrium-aluminum garnet " Nd:YAG " lasers).
- The neodymium-Yag laser emits infrared light at 1,064 nanometers ($\text{nm} = 1 \times 10^{-9}$ meters).
- They are high power lasers and used for industrial applications such as welding, drilling, cutting, molding etc.
- These applications require very high power with peak value in kilowatts to megawatts.
- solid state lasers use high density solid media as active laser materials

Examples of solid state laser types are **ruby laser, Nd:YAG laser** etc.

- **Gas lasers** : They are widely available for all power (mwatts to Mwatts) and wavelength (UV, IR) requirements.
- It uses low density gaseous materials as active media.
- Electrical pumping (continuous, RF or pulsed) is used.
- helium and helium-neon, He-Ne, are the most common gas lasers
- These have a primary output of visible red light.
- CO₂ lasers emit energy in the far-infrared, and are used for cutting hard materials
- Gas lasers can be made from neutral atoms (He-Ne, metal vapor etc.), ions (e.g. Ar⁺) or molecules (e.g. CO₂)

Examples of gas laser types are argon laser, CO₂ laser, He-Ne laser etc.

- **Liquid laser or Dye laser:** Liquid lasers use liquid as active medium.
- In dye laser, liquid material is called dye (e.g. rhodamine B, sodium fluorescein, rhodamine 6G) is used as an active medium, which causes to produce laser light.
- These lasers produce output whose wavelengths are in visible, ultra-violet and near infrared spectrum.
- It is used as research tool in medical applications.

- **Excimer lasers** (the name is derived from the terms *excited* and *dimers*) use reactive gases, such as chlorine and fluorine, mixed with inert gases such as argon, krypton or xenon.
- When electrically stimulated, a pseudo molecule (dimer) is produced.
- When lased, the dimer produces light in the ultraviolet range.

- **Semiconductor lasers** : Semiconductor lasers are compact in size as they are made using semiconductor materials on nanometer scale accuracy.
- It is similar to transistor and has operation like LED, but the output beam has characteristics of laser light.
- The material most often used in semiconductor laser is GaAs (Gallium Arsenide).
- Hence it is known as gallium arsenide laser. It is also called as injection laser

Examples of semiconductor lasers are Homojunction laser, double heterojunction laser , Quantum well laser, distributed feedback laser, tunable laser, surface emitting laser etc

- **Semiconductor lasers, sometimes called diode lasers,** are not solid-state lasers.
- These electronic devices are generally very small and use low power.
- They may be built into larger arrays, such as the writing source in some laser printers or CD players

Spatial and Temporal Coherence

- Coherence is one of the most important concepts in optics and is strongly related to the ability of light to exhibit interference effects.
- A light field is called *coherent* when there is a fixed phase relationship between the electric field values at different locations or at different times.

There are two very different aspects of coherence:

Spatial coherence

Temporal coherence

Spatial coherence

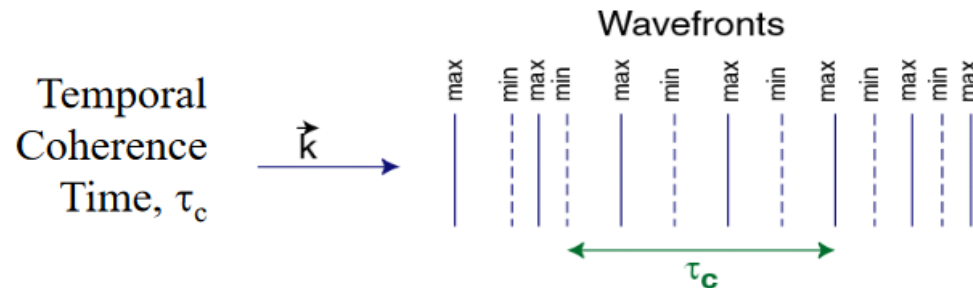
- Spatial coherence means a strong correlation (fixed phase relationship) between the electric fields at **different locations across the beam profile**.
- Spatial Coherence is a measure of the correlation of a light wave's phase at different points **transverse to the direction of propagation** - it tells us how uniform the phase of a wavefront is
- **Spatial coherence is the essential prerequisite of the strong directionality of laser beams.**

Temporal coherence

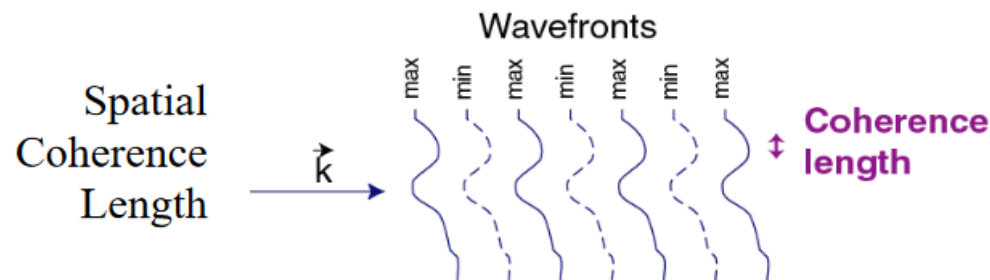
- Temporal coherence means a strong correlation between the electric fields at **one location but different times**.
- Temporal Coherence is a measure of the correlation of light wave's phase at different points **along the direction of propagation** – it tells us how monochromatic a source is.
- For example, the output of a single-frequency laser can exhibit a very **high temporal coherence**, as the electric field temporally evolves in a highly predictable fashion: it exhibits a clean sinusoidal oscillation over extended periods of time.

The Temporal Coherence Time and the Spatial Coherence Length

The temporal coherence time is the time the wave-fronts remain equally spaced. That is, the field remains sinusoidal with one wavelength:



The spatial coherence length is the distance over which the beam wave-fronts remain flat:



Since there are two transverse dimensions, we can define a coherence area.

The coherence time is given by:

$$\tau_c = 1 / \Delta\nu$$

where $\Delta\nu$ is the light bandwidth (the width of the spectrum).

Sunlight is temporally very incoherent because its bandwidth is very large (the entire visible spectrum).

Lasers can have coherence times as long as about a second, which is amazing; that's $>10^{14}$ cycles!

Applications of Lasers

- The most significant applications of lasers include:
- in medicine
- in communications
- in industries
- in science and technology
- in military

Lasers in Medicine

- Lasers are used for bloodless surgery.
- Lasers are used to destroy kidney stones.
- Lasers are used in cancer diagnosis and therapy.
- Lasers are used for eye lens curvature corrections.
- Lasers are used in fiber-optic endoscope to detect ulcers in the intestines.
- The liver and lung diseases could be treated by using lasers.
- Lasers are used to study the internal structure of microorganisms and cells.
- Lasers are used to produce chemical reactions.
- Lasers are used to create plasma.
- Lasers are used to remove tumors successfully.
- Lasers are used to remove the caries or decayed portion of the teeth.
- Lasers are used in cosmetic treatments such as acne treatment, cellulite and hair removal.

Ophthalmology

- Historically, **ophthalmology was the first medical field where lasers** have found an application.
- Over more than five decades, use of lasers in ophthalmology has successfully shown effective and safe results in treating various eye conditions.
- lasers are used to correct vision or repair damage due to degenerative diseases dictates optimal laser type, wavelength, and pulse length.

Refractive Eye Surgery

- Since the introduction of lasers in refractive eye surgery, technology has continually evolved to refine the technique and minimize the risk of damage to the retina and optic nerve.
- **Femtosecond lasers** are used in conjunction with other technology in LASIK (laser in situ keratomileusis) procedures to correct myopia (nearsightedness).

Cataract Treatment

- Cataracts, or clouding of the eye's natural lens, are the most common cause of vision loss in the elderly population.
- Accordingly, cataract surgery is the most commonly performed surgical procedure in the world, Treatment involves removal of the lens for a prosthetic replacement.
- Lasers have recently been developed as a means to remove the lens via photodisruption.
- **Femtosecond lasers** are used in the treatment of cataracts and take advantage of the extremely high peak power densities to efficiently disrupt tissue with minimal surrounding thermal damage.

Retinal (Micro) Coagulation

- Diabetic macular edema is the most common cause of visual loss in persons under 50 years of age in the developed world.
- Retinal photocoagulation has been developed and recognized as a standard of treatment.
- Recently, the procedure was perfected due to introduction of micro-coagulation technique employing pulsed diode lasers.

Thank you